ONE COEFFICIENT, ONE DIRECTION

The graph of $P(x) = ax + b$ is a single straight line with no curvature and no turning points. The coefficient a governs its direction and steepness.

LINEAR POLYNOMIAL

$$P(x) = ax + b$$

split by **the sign of a**

↗

CASE 1

$a > 0$

Climbs from left to right

increases everywhere and never changes direction

↘

CASE 2

$a < 0$

Descends from left to right

decreases everywhere and never changes direction

TAKEAWAY

Two points fix the whole graph: $(0, b)$ on the vertical axis and the root at $x = -\dfrac{b}{a}$. Doubling a doubles the steepness.